Physics 5A Midterm 1 Review Session

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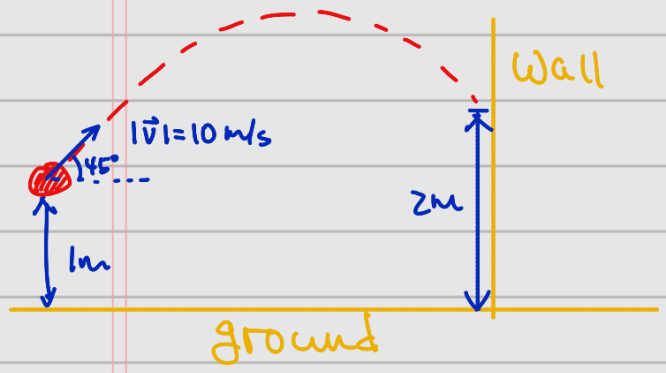
Problem 1: During the discussion last week, Jinyan threw a stress ball against the wall. The ball left Jinyan's hand at 1 meter above the ground with an initial velocity 10 m/s at an angle 45° above the horizon. The ball reach its maximum height before hitting the wall, and it hit the wall at a height 2 meters above the ground.

Part 1: Find x- and y-components of the ball's initial velocity.
Part 2: Find the time it takes for the ball to hit the wall.
Part 3: Find the total horizontal distance it has traveled before reaching the wall.
Part 4: Find the maximal height the ball could reach.
Part 5: Find the ball's final velocity when it hit the wall (write it as $\vec{v} = (v_x, v_y)$).

Extra parts that you should know how to calculate:
Part 6: Find the total vertical displacement of the ball in its journey.
Part 7: Find the total vertical distance traveled in its journey.
Part 8: Find the total displacement vector and average velocity.

Problem 2: Now instead of throwing the ball at an angle 45° above the horizon, Jinyan throws the ball directly upward. The ball again starts at 1 meter above the ground with an initial velocity 10 m/s. The ball then drops to the ground after a few seconds.

Part 1: Find the time it takes for the ball to hit the ground.
Part 2: Find the maximal height the ball could reach.
Part 3: Find the total vertical distance it has traveled before hitting the ground.
Part 4: Find the average speed of the ball in its journey.
Part 5: Find the total displacement vector and average velocity.



① setup x and y axes (pick a "reasonable one")

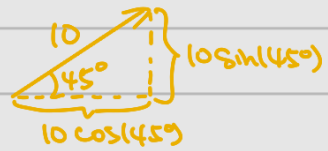
We pick the origin here

You can also pick the origin at where the initial position of the ball is, just make sure being consistant.

② After the origin is fixed we can write down

$\vec{x}_0 = (0\text{m}, 1\text{m})$ initial position

$\vec{V}_0 = (10\cos(45°)\text{ m/s}, 10\sin(45°)\text{ m/s})$ initial velocity b/c



$\vec{a} = (0\text{ m/s}^2, -9.8\text{ m/s}^2)$

constant acceleration by gravity. No other thing is trying to make the ball move (no boost equipped, no one is pushing / pulling the ball) so only acceleration is by gravity directly downward

$\downarrow \vec{a}$

③ Use kinematic eqns

describe motion in x-direction & y-direction separately, independently

$x(\text{at time } t) = \frac{1}{2}a_x t^2 + V_{0,x}t + x_0$

$y(\text{at time } t) = \frac{1}{2}a_y t^2 + V_{0,y}t + y_0$

ball's y-position at time t

$x = \frac{1}{2}\,0\,t^2 + 10\cos(45°)t + 0$

$y = \frac{1}{2}(-9.8)t^2 + 10\sin(45°)t + 1$

Now plug in #'s that we know

$$\begin{cases} \vec{a} = (a_x, a_y) = (0, -9.8) \\ \vec{V}_0 = (V_{0,x}, V_{0,y}) = (10\cos(45°), 10\sin(45°)) \\ \vec{x}_0 = (0, 1) \end{cases}$$

④ Now look at the questions. Done!

Part 1. Find initial velocity components $\vec{V}_0 = (10\cos(45°)\ m/s,\ 10\sin(45°)\ m/s)$

Part 2 the ball hit the wall at $y=2m$

$y = \frac{1}{2}(-9.8)t^2 + 10\sin(45°)t + 1$ ← use this, set $y=2$

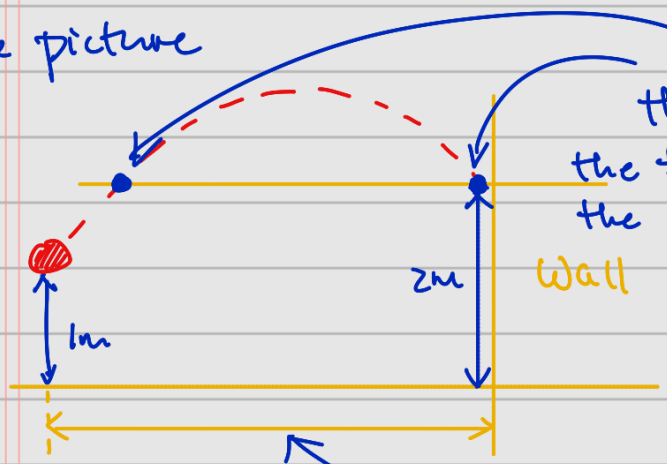
so $2 = -4.9t^2 + 5\sqrt{2}t + 1 \Rightarrow 0 = \underbrace{-4.5}_{a}t^2 + \underbrace{5\sqrt{2}}_{b}t \underbrace{-1}_{c}$

use quadratic formula $t = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = 0.159$ or 1.284

We get two #'s for t such that $y=2\ m$

Wut?

use the picture



the ball reaches $y=2$ twice

the first time happens @ $t=0.159$

the second time @ $t=1.284$

this is the answer we want to find

Part 3 find this distance

We know the ball travelled $t=1.284s$ before hitting the wall

$x = \frac{1}{2}0t^2 + 10\cos(45°)t + 0$ ← plug it in to find x

so $x = 10\cos(45°)\cdot 1.284 = 9.07$ meters.

so the ball traveled 9.07 m before hitting the wall.

Part 4 Maximal height

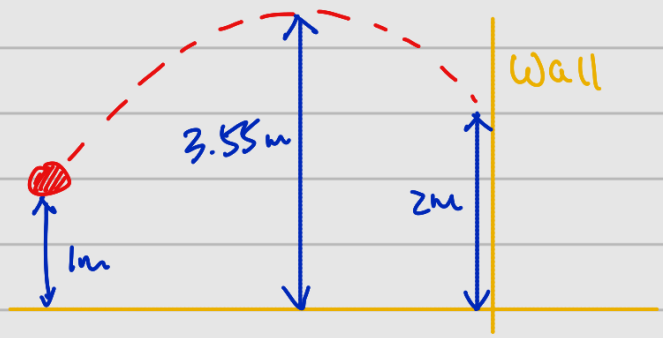
$y = \frac{1}{2}(-9.8)t^2 + 10\sin(45°)t + 1$ ← use this, but find largest y.

for parabola $y = at^2 + bt + c$ y is maximized when $t = -\frac{b}{2a}$

so at $t = \frac{10\sin(45°)}{-2(-9.8/2)} = 0.722\ s$ y is maximized

$y_{max} = \frac{1}{2}(-9.8)(0.722)^2 + 10\sin(45°)\cdot 0.722 + 1$

$= 3.55\ m.$



Part 4 Alternatively.

$V_y = 0$ when the ball reaches maximal height

"y-component" of ball's velocity.

this is b/c the ball "turns around" in the vertical direction when at maximum height

$V_y = a_y t + V_{0,y}$ set $V_y = 0$

so $0 = -9.8t + 10\sin(45°)$ $\xrightarrow{\text{get } t \text{ for max height}}$ $t = \dfrac{10\sin(45°)}{9.8}$

Same as above !!

But this understanding is important for part 5

Part 5 * Recall the ball hits the wall at $t = 1.284$ s.

* Gravity changes y-component of $\vec{v}$
* Nothing changes x-component of $\vec{v}$

so V_y at time $t = 1.284$ s

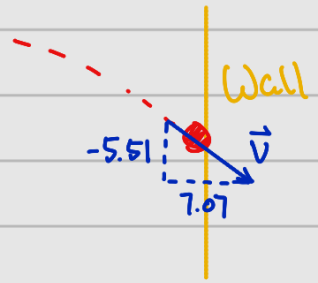
$V_y = a_y t + V_{0,y} = (-9.8)\cdot 1.284 + 10\sin(45°) = -5.51$ m/s

V_x at time $t = 1.284$ s

$V_x = \cancel{a_x}^{\,0} t + V_{0,x} = 10\cos(45°) = 7.07$ m/s same as $V_{0,x}$

so at the time the ball hits the wall

$\vec{V} = (7.07 \text{ m/s}, -5.51 \text{ m/s})$



* magnitude $|\vec{v}| = \sqrt{7.07^2 + 5.51^2}$
* angle $\theta = \tan^{-1}\left(\dfrac{-5.51}{7.07}\right)$

Part 6 total vertical displacement

$\Delta y = y_{final} - y_{initial} = 2m - 1m = 1m$

So the ball goes up 1 m in total in its journey.

Part 7 total vertical distance traveled

* first go from $y = 1$ m to $y_{max} = 3.55$ m $\Rightarrow |\Delta y| = |3.55 - 1|$ m.
* then "turn around" in vertical direction
* then go from $y_{max} = 3.55$ m to $y = 2$ m $\Rightarrow |\Delta y| = |2 - 3.55|$ m

Sum $\Rightarrow$ total vertical distance $|3.55 - 1| + |2 - 3.55| = 4.10$ m.

Part 8 total displacement

final x-position, final y-position, displacement in x-direction

$$\Delta\vec{x} = \vec{x}_{final} - \vec{x}_{initial} = (9.07\,m,\ 2\,m) - (0\,m,\ 1\,m) = (9.07\,m,\ 1\,m)$$

initial x and y positions.

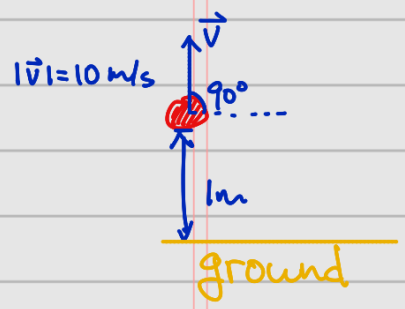
total displacement

$$\text{average velocity} = \frac{\Delta\vec{x}}{\Delta t} = \left(\frac{9.07\,m}{1.28\,s},\ \frac{1\,m}{1.28\,s}\right) = (7.08\,m/s,\ 0.78\,m/s)$$

total time it takes to hit the wall from part 2.

y-component of avg. velocity

Now throw the ball directly upward



Same procedure as above

$$\begin{cases} \vec{x}_0 = (0\,m,\ 1\,m) \\ \vec{v}_0 = (10\cos(90°)\,m/s,\ 10\sin(90°)\,m/s) \\ \vec{a} = (0\,m/s^2,\ -9.8\,m/s^2) \end{cases}$$

plug in #'s

$$x(\text{at time } t) = \tfrac{1}{2}a_x t^2 + v_{0,x}t + x_0 \qquad \Rightarrow \qquad x = \tfrac{1}{2}\,0\,t^2 + 10\cos(90°)t + 0$$

$$y(\text{at time } t) = \tfrac{1}{2}a_y t^2 + v_{0,y}t + y_0 \qquad\qquad y = \tfrac{1}{2}(-9.8)t^2 + 10\sin(90°)t + 1$$

Notice $x = \frac{1}{2}\cdot 0\,t^2 + 10\,\cancel{\cos(90°)}^{\,0}\,t + 0 = \underline{\underline{0}}$

So ball doesn't move horizontally. it stays at $x=0$ for x-position.

Ofc. b/c we throw it directly upward.

So $y = \frac{1}{2}(-9.8)t^2 + 10\sin(90°)t + 1$ tells the full story

Part 1. hit the ground? set $y=0$

$0 = -4.9t^2 + 10t + 1 \Rightarrow t = 2.13\,s$ and a negative t which we disregard b/c it "happens" before we throw the ball

it takes $t = 2.13\,s$ to hit the ground

Part 2 Maximal height? use $t = -\frac{b}{2a}$ as in problem 1.

So at $t = \frac{10\sin(90°)}{-2(-9.8/2)} = 1.02\,s$ y is maximized

$y_{max} = -4.9(1.02)^2 + 10(1.02) + 1 = 6.10\,m.$

Part 3. total vertical distance traveled

* first go from $y = 1\,m$ to $y_{max} = 6.10\,m \Rightarrow |\Delta y| = |6.10 - 1|\,m$.
* then "turn around" in vertical direction
* then go from $y_{max} = 6.10\,m$ to $y = 0\,m \Rightarrow |\Delta y| = |0 - 6.10|\,m$

Sum $\Rightarrow$ total vertical distance $|6.10-1| + |0-6.10| = 11.20\,m$.

Part 4 average speed in vertical direction $= \dfrac{\text{total vertical distance}}{\text{total time it takes}}$.

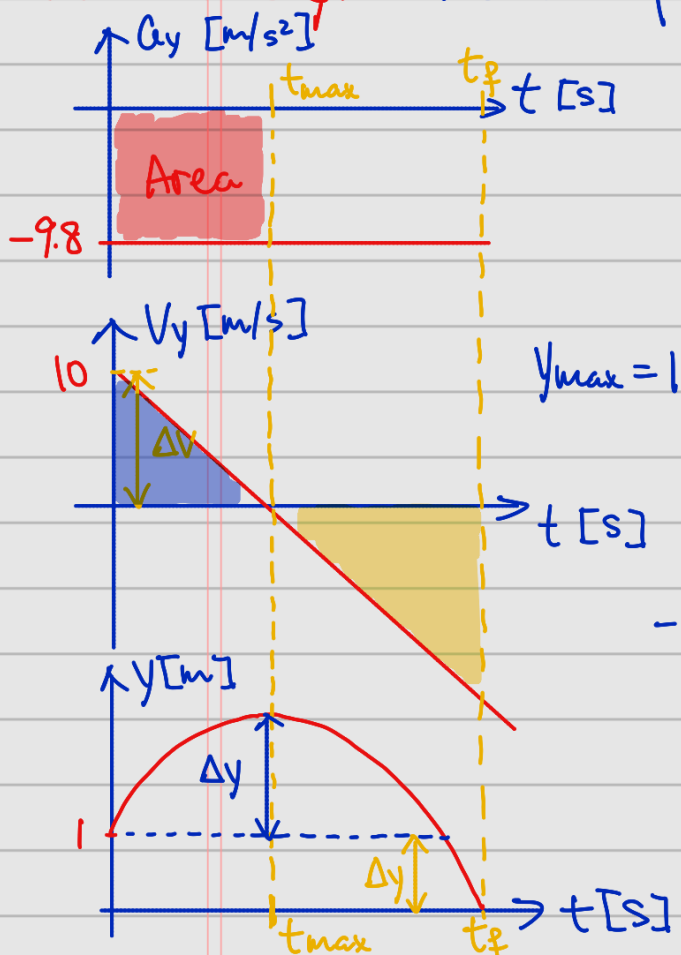
$$\bar{s} = \frac{11.2\,m}{\underbrace{2.13\,s}_{\text{from part 1.}}} = 5.258\,m/s.$$

Part 5 total displacement

$$\Delta y = y_{final} - y_{initial} = 0\,m - 1\,m = -1\,m$$

$$\text{average velocity} = \frac{\overbrace{\Delta y}^{\text{total displacement}}}{\underbrace{\Delta t}_{\text{total time it takes to hit the ground}}} = \frac{-1\,m}{2.13\,s} = -0.47\,s.$$

Alternatively. solve the problem using graphs



find t_{max} (for y_{max})

$$-10 = \Delta V = \text{Area} = t_{max} \cdot (-9.8)$$

$$\Rightarrow t_{max} = 1.02\,s \quad \text{part 2}$$

then find y_{max}

$$y_{max} = 1 + \Delta y = 1 + \text{Area} = 1 + (1.02 \cdot 10)/2 = 6.10\,m$$

find t_f (ball hits the ground)

$$(0-1) = -1 = \Delta y = \text{Area} + \text{Area}$$

$$-1 = (1.02 \cdot 10)/2 + (t_f - 1.02)(t_f - 1.02)(-9.8)/2$$

$$\hookrightarrow t_f = 2.13\,s \quad \text{part 1}$$